

Business Requirements Document (BRD)

Hotel Booking Agentic Chatbot — Backend

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Status: Draft

1. Business Problem

Hotels and booking desks spend significant staff time answering repetitive guest queries: room availability, pricing, amenity details, and booking confirmations. These interactions are largely formulaic but require 24/7 availability. An autonomous agentic backend can handle the full booking conversation without human intervention.

2. Business Objectives

- 1 Automate end-to-end hotel booking conversations for a single property.
- 2 Reduce time-to-confirmation for a guest query from minutes (human staff) to seconds (agent).
- 3 Demonstrate a production-grade agentic workflow pattern that interns can study, extend, and test.

3. Scope

In Scope:

Natural language understanding of booking intent (check-in/check-out dates, room type, guests).

Availability checking against a mock hotel inventory.

Booking creation and confirmation generation.

FAQ handling (amenities, cancellation policy, check-in times, pet policy).

Multi-turn conversation management within a single booking session.

Out of Scope:

Real payment processing or PCI-DSS compliance.

Frontend / UI / REST API layer.

Multi-property or OTA (Online Travel Agency) integrations.

User account management or authentication.

4. Stakeholders

Role	Expectation
Hotel Operations	Receive structured booking records; no manual transcription.
Guest (simulated)	Fast, accurate answers; clear confirmation.
Intern / Engineer	Readable agentic pattern they can instrument and extend.

5. High-Level Requirements

ID	Requirement
BR-01	The system must understand a guest's booking intent from natural language input.
BR-02	The system must check availability for the requested dates and room type.
BR-03	The system must create a booking record when the guest confirms.
BR-04	The system must answer common hotel FAQ questions without escalating.
BR-05	The system must produce a structured booking confirmation at the end of each session.
BR-06	The system must handle ambiguous or incomplete requests by asking targeted clarifying questions.
BR-07	The system must escalate gracefully (produce a handoff record) if it cannot resolve a query within configured retry limits.

6. Success Metrics

Booking intents correctly classified: $\geq 95\%$ on the sample test set.

Average turns to booking confirmation: ≤ 5 .

FAQ queries resolved without clarification: $\geq 90\%$.

No unhandled exceptions during a full booking conversation run.